

Burns & Plastic Surgery — Examination Master Guide

Complete 20-Mark Master Solutions: Pathophysiology, Surgical Ladders, Grafts, Flaps, Aesthetics & Microsurgery

Coverage: 6 Comprehensive Topics | Burns Management, Reconstructive Surgery, Skin Grafts, Cosmetic Surgery, Scars & Contractures, Microvascular Surgery.

1. Burns: Classification, Pathophysiology, and Comprehensive Management

20 MARKS TOPIC

Definition: A burn is an acute traumatic injury to the skin, subcutaneous tissues, or deeper structures caused by thermal energy (flame, scald, contact), chemical agents, electricity, or radiation, resulting in protein denaturation, cellular necrosis, and systemic burn shock.

Classification by Depth of Injury

Depth Category	Anatomical Level	Clinical Appearance	Sensation	Healing / Intervention
1st Degree (Superficial)	Epidermis only	Erythematous, dry, no blisters	Hyperesthetic, painful	Heals in 3–6 days; no scarring
2nd Degree (Superficial Partial)	Epidermis + Papillary Dermis	Pink/red, moist, thin-walled bullae; brisk refill	Intensely painful	Heals in 7–14 days; minimal scar
2nd Degree (Deep Partial)	Epidermis + Reticular Dermis	Mottled white/red, waxy, dry; sluggish refill	Dull pinprick sensation	Heals in 3–8 weeks with hypertrophic scar; needs excision & grafting
3rd Degree (Full-Thickness)	Entire Epidermis + Dermis	Leathery, porcelain white/charred; non-blanching	Insensate (anesthetic)	Requires surgical excision & skin grafting
4th Degree (Subdermal)	Extends into fat, fascia, muscle, bone	Charred, blackened, exposed bone/muscle	Anesthetic	Debridement, flap coverage, or amputation

Pathophysiology of Burn Injury

- **Jackson's Zones of Burn Injury:**
 - *Zone of Coagulation (Central):* Point of maximum damage; irreversible protein denaturation and tissue necrosis.
 - *Zone of Stasis (Middle):* Compromised tissue perfusion and ischemia; **salvageable with adequate fluid resuscitation**; prone to secondary necrosis if untreated.
 - *Zone of Hyperemia (Outer):* Peripheral vasodilation due to inflammatory mediator release; viable tissue that recovers spontaneously.
- **Systemic Burn Shock & SIRS (TBSA > 20–25%):** Widespread release of histamine, bradykinin, prostaglandins, and oxygen radicals causes massive microvascular hyperpermeability. Intravascular fluid and albumin leak into the interstitium, causing profound hypovolemia, hemoconcentration, and hypermetabolism (resting energy expenditure increases up to 200%).

Burn Size Estimation (TBSA)

- **Wallace Rule of Nines (Adults):** Head & Neck = 9%, Each Arm = 9% (18% total), Anterior Trunk = 18%, Posterior Trunk = 18%, Each Leg = 18% (36% total), Perineum = 1%.
- **Lund-Browder Chart:** Gold standard for pediatrics (accounts for disproportionately large head and smaller limbs in children).
- **Palmar Surface Rule:** Patient's open hand (palm + fingers) $\approx$ 1% TBSA.

Acute Resuscitation Protocol

Parkland (Baxter) Formula:

Total 24-hour Fluid (Ringer's Lactate) = $4 \text{ mL} \times \text{Body Weight (kg)} \times \% \text{TBSA}$

- **First 8 Hours:** Administer 50% of the calculated fluid (measured from the *exact time of burn injury*).
- **Next 16 Hours:** Administer the remaining 50% of the total fluid volume.

Updated ABA Consensus: Recommends 2 mL/kg/%TBSA for adult thermal burns to avoid over-resuscitation ("fluid creep").

Monitoring Endpoints: Target hourly urine output of **0.5 mL/kg/hr** in adults (30–50 mL/hr) and **1.0 mL/kg/hr** in children. For electrical burns with myoglobinuria, maintain 1.5–2.0 mL/kg/hr until clear.

Surgical Management

- **Emergency Escharotomy:** Indicated for circumferential full-thickness extremity burns with compromised distal pulses, or torso burns restricting respiratory excursions. Involves longitudinal incision through the rigid eschar into subcutaneous fat.
- **Early Tangential Excision:** Performed within Days 2–5 post-stabilization; horizontal shaving of necrotic eschar down to healthy punctate bleeding tissue followed by immediate split-thickness skin grafting (STSG).

2. Reconstructive Surgery and The Reconstructive Ladder

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Definition: Reconstructive surgery is a surgical discipline focused on restoring normal anatomical form, function, and aesthetic contour to tissues deformed or lost secondary to trauma, congenital anomalies, oncologic resection, or burn sequelae.

The Reconstructive Ladder vs. Reconstructive Elevator

[RECONSTRUCTIVE LADDER]

Level 6: Free Tissue Transfer (Microvascular)
Level 5: Distant / Regional Pedicled Flaps
Level 4: Local Tissue Flaps (Z-plasty, Adv)
Level 3: Full-Thickness Skin Grafts (FTSG)
Level 2: Split-Thickness Skin Grafts (STSG)
Level 1: Secondary Intention / Primary Closure

[RECONSTRUCTIVE ELEVATOR]

→ Choose the technique providing the BEST functional & aesthetic result, not necessarily the simplest. (e.g., exposed bare bone/tendon requires an immediate flap).

Surgical Flap Classification

- **By Blood Supply:**
 - *Random Pattern Flaps:* Supplied by non-directional microvascular subdermal plexus; strict length-to-width ratio (1:1 to 1.5:1).
 - *Axial Pattern Flaps:* Supplied by a well-defined, named anatomical arteriovenous system running along the long axis (e.g., Groin flap, Deltopectoral flap).

- **Perforator Flaps:** Supplied by isolated perforating vessels traversing deep fascia/muscle, allowing flap harvest while preserving underlying muscle (e.g., DIEP, ALT).

- **By Tissue Composition:** Cutaneous, Fasciocutaneous (skin + fat + deep fascia), Myocutaneous (muscle + skin paddle), and Osteocutaneous (bone + soft tissue).
- **By Flap Movement:** Advancement (V-Y, rectangular), Transposition (Limberg rhomboid, Z-plasty), Rotation (semicircular arc), and Interpolation (pedicle crosses intact skin, e.g., Paramedian Forehead flap).

Mathes & Nahai Classification of Muscle Flaps

Type	Vascular Anatomy	Dominant Pedicles	Minor / Secondary	Clinical Examples
Type I	Single vascular pedicle	One single artery/vein	None	Tensor Fasciae Latae (TFL), Gastrocnemius
Type II	Dominant + Minor pedicles	Dominant supplies muscle	Minor enters distal base	Gracilis, Trapezius, Soleus
Type III	Two codominant pedicles	Both major (either sustains)	None	Rectus Abdominis, Gluteus Maximus
Type IV	Segmental pedicles	No single dominant	Multiple along length	Sartorius, Tibialis Anterior
Type V	1 Dominant + Segmentals	One large dominant supply	Segmentals support base	Latissimus Dorsi, Pectoralis Major

3. Skin Grafts

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Definition: A skin graft is a segment of avascular, completely detached epidermis and dermis harvested from a donor site and transferred to a recipient vascular bed, relying entirely on host bed neovascularization for survival.

Classification: STSG vs. FTSG

Parameter	Split-Thickness Skin Graft (STSG)	Full-Thickness Skin Graft (FTSG)
Composition	Epidermis + Variable portion of Dermis (0.008–0.018 in)	Epidermis + Entire Dermis (> 0.020 in)
Donor Site Healing	Re-epithelializes spontaneously from adnexa (follicles)	Requires primary surgical closure or local flap
Primary Contraction	Minimal (~10% immediate elastic recoil)	High (~40–50% recoil upon harvest)
Secondary Contraction	High (myofibroblast contracture of recipient bed)	Minimal (thick dermis resists wound bed contracture)
Graft "Take" / Survival	Rapid, highly reliable; tolerates less vascular beds	Slower; requires pristine, highly vascularized bed
Aesthetic Match	Inferior (shiny, pigmentary changes, contour dip)	Superior (matches color, texture, and hair)
Primary Indications	Extensive burns, large traumatic wounds, fasciotomies	Facial defects, eyelids, hands, aesthetic units

Stages of Skin Graft Healing ("Graft Take")

- 1 **Plasmatic Imbibition (Hours 0–48):** The avascular graft acts like a sponge, passively absorbing nutrients, electrolytes, and oxygen from host wound exudate by capillary action; fibrin acts as temporary anchor. Graft weight increases by 30–40%.
- 2 **Inosculation (Days 2–4):** Direct end-to-end alignment and micro-anastomoses ("kissing vessels") establish connection between graft microvessels and budding host capillaries.
- 3 **Revascularization & Angiogenesis (Day 4 onward):** Host endothelial capillary ingrowth penetrates the graft dermis, establishing mature arterial inflow, venous outflow, and lymphatic drainage. Fibroblasts deposit permanent anchoring collagen.

Causes of Graft Failure

1. **Hematoma / Seroma:** Most common cause; fluid elevates graft from the bed, disrupting imbibition and inosculation.
2. **Shearing Forces:** Motion severs fragile microvessels.
3. **Infection:** Bioburden $> 10^5$ organisms/g or *Group A Streptococcus* (secretes streptokinase, dissolving the fibrin bond).
4. **Inadequate Bed:** Exposed avascular bone lacking periosteum, tendon without paratenon, or cartilage without perichondrium.

4. Cosmetic (Aesthetic) Surgery

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Definition: Cosmetic surgery is an elective subspecialty of plastic surgery directed toward enhancing, restoring, and harmonizing normal anatomical structures to optimize physical appearance, facial balance, and patient self-perception.

Core Aesthetic Surgical Procedures

- **1. Rhytidectomy (Facelift):**
 - *Anatomy:* Focuses on repositioning the **SMAS (Superficial Musculoaponeurotic System)**, a fibromuscular layer continuous with platysma and temporoparietal fascia.
 - *Techniques:* SMAS plication/imbrication (suturing/resecting SMAS redundancy) vs. *Deep-Plane Facelift* (dissecting beneath SMAS, releasing zygomatic and masseteric retaining ligaments to suspend midface and jowls without skin tension).
- **2. Blepharoplasty (Eyelid Rejuvenation):**
 - *Upper Blepharoplasty:* Excision of redundant upper lid skin (**dermatochalasis**) and conservative resection of herniated medial/central fat compartments.
 - *Lower Blepharoplasty:* Subciliary or transconjunctival approach; addresses lower lid steatoblepharon (bags) via fat resection or fat transposition across the tear trough (*arcus marginalis release*).
- **3. Rhinoplasty:**
 - *Approaches:* Closed (endonasal incisions) vs. Open (trans-columellar inverted-V incision providing wide structural exposure).
 - *Modifications:* Dorsal hump resection, osteotomies to close open-roof deformity, spreader grafts (to maintain internal nasal valve angle $> 10\text{--}15^\circ$), and columellar strut grafts for tip support.
- **4. Abdominoplasty ("Tummy Tuck"):**
 - Indicated for postpartum or massive weight loss abdominal wall laxity.
 - *Operative Steps:* Low transverse suprapubic incision → Elevation of dermoadipose flap to costal margins → **Plication of rectus abdominis anterior fascia (corrects rectus diastasis)** → Transposition of umbilicus → Resection of redundant skin apron.

- **5. Liposuction:** Suction-Assisted (SAL), Ultrasound-Assisted (UAL), or Power-Assisted (PAL). Involves subcutaneous infiltration of ****Klein's Tumescence Solution**** (Lactated Ringer's + dilute Epinephrine 1:1,000,000 + Lignocaine) to induce vasoconstriction, minimizing blood loss and providing regional anesthesia.

5. Scar Management and Contracture Correction

20 MARKS TOPIC

Definition: *Scar Management* encompasses prophylactic and therapeutic strategies to optimize wound remodeling. *Contracture* is the pathological, rigid shortening of mature scar tissue across joints or orifices resulting in functional limitation and anatomical distortion.

Hypertrophic Scar vs. Keloid

Feature	Hypertrophic Scar	Keloid
Growth Extent	Confined strictly within original wound margins	Invades beyond original wound boundaries into normal skin
Etiology & Genetics	Related to wound tension, infection, or delayed healing	Strong genetic predisposition; common in darker skin (Fitzpatrick IV–VI)
Onset & Course	Develops early (within 4 weeks); often regresses over 1–2 yrs	Develops late (months/years after minor trauma); progressive growth
Common Sites	Flexor surfaces, joints, areas under mechanical tension	Sternal presternal region, earlobes, deltoid, upper back, jawline
Histology	Organized parallel bundles of Type III collagen	Disorganized, thick, whorled glassy Type I & III collagen
Surgical Recurrence	Low recurrence after tension-free revision	Very high recurrence (45–100%) if excised without adjuvants

Therapeutic Scar Management Spectrum

- **Non-Surgical Measures:**
 - *Silicone Gel Sheeting:* First-line standard; increases stratum corneum hydration and downregulates cytokine-mediated fibroblast hyperactivity.
 - *Pressure Therapy:* Custom elastic garments applying 15–25 mmHg continuous pressure for 23 hours/day.
 - *Intralesional Injections:* **Triamcinolone Acetonide (10–40 mg/mL)** ± 5-Fluorouracil (5-FU) at 4–6 week intervals (inhibits fibroblast proliferation and collagen synthesis).
 - *Laser Therapy:* 595 nm Pulsed Dye Laser (PDL) to ablate microvascular feeding vessels.

Surgical Correction of Contractures

- **Z-Plasty Principles & Angles:** A double-transposition flap technique that lengthens a contracted scar along its central limb and reorients scar direction into relaxed skin tension lines (RSTL).
 - **30°–30° Z-plasty:** 25% theoretical gain in length.
 - **45°–45° Z-plasty:** 50% theoretical gain in length.
 - **60°–60° Z-plasty (Classic Standard):** 75% theoretical gain in length with a 90° change in scar axis.
- **W-Plasty & Geometric Broken Line Closure:** Breaks up long linear facial scars into irregular zig-zag interdigitations to diffuse light and camouflage scars into RSTLs without changing length.

6. Microvascular Surgery and Free Tissue Transfer

20 MARKS TOPIC

Definition: Microvascular surgery is a specialized operative discipline utilizing high-power optical magnification (operating microscopes), specialized micro-instruments, and micro-sutures (8-0 to 11-0) to perform vascular anastomoses on small vessels (0.5–3.0 mm) and coaptation of peripheral nerve fascicles.

Surgical Technique: Microvascular Anastomosis

- 1 Vessel Preparation:** Mobilize donor and recipient vessels under the microscope; circumferentially strip perivascular adventitia back 2–3 mm from cut edges to prevent thrombogenic adventitial prolapse into the lumen.
- 2 Mechanical Dilation & Flushing:** Gently dilate vessel ends with jewelers forceps; irrigate lumen with heparinized saline (100 IU/mL) to clear debris and microthrombi.
- 3 Bi-triangulation / Triangulation:** Place two stay sutures at 180° (or three at 120°) using 9-0 or 10-0 nylon. Keep stay sutures under gentle traction to keep anterior and posterior vessel walls separated.
- 4 Suture Placement:** Place interrupted through-and-through sutures across adventitia, media, and intima, ensuring precise intima-to-intima eversion without catching the opposite ("back") wall.
- 5 Reperfusion:** Release the venous clamp first, followed by the arterial inflow clamp. Confirm patency using the **Acland micro-forceps strip test**.

Common Workhorse Free Flaps

Flap Name	Tissue Harvested	Vascular Pedicle	Major Clinical Applications
Anterolateral Thigh (ALT)	Fasciocutaneous / Myocutaneous	Descending branch of Lateral Circumflex Femoral Artery	Head & neck reconstruction, oral cavity, soft tissue defects
Free Fibula Flap	Osteocutaneous (Vascularized bone)	Peroneal Artery and Venae Comitantes	Mandibular / Maxillary reconstruction , long bone defects
Radial Forearm (RFFF)	Thin, pliable Fasciocutaneous	Radial Artery and Venae Comitantes / Cephalic vein	Intraoral lining (tongue, floor of mouth), pharynx
DIEP Flap	Skin & Subcutaneous Fat (Spare muscle)	Deep Inferior Epigastric Perforators	Autologous post-mastectomy breast reconstruction
Latissimus Dorsi (LD)	Muscle / Myocutaneous	Thoracodorsal Artery and Vein	Large soft-tissue coverage, chest wall, scalp defects

Postoperative Monitoring & Flap Salvage

- **Clinical Parameters (Checked hourly for first 24 hrs):**
 - **Color & CRT:** Healthy pink with 1–2 sec refill vs. Pale (arterial thrombosis) vs. Violaceous/Congested with < 1 sec rapid refill (venous thrombosis).
 - **Temperature:** Monitored via surface probe (drop > 2–3°C indicates perfusion compromise).
 - **Pinprick Bleeding:** Bright red within 1–2 sec is normal; absent bleeding indicates arterial occlusion; dark purple rapid bleeding indicates venous outflow obstruction.
- **Emergency Flap Salvage:** Immediate return to theatre within 4–6 hours of suspected compromise. Re-exploration involves evacuating compressive hematomas, topical vasodilators (Papaverine), catheter-directed thrombolysis (tPA), re-doing anastomoses, or interposition vein grafting.